



DM-OLED139-654-24Pin

1.39" 400 × 400 AMOLED ROUND
FULL COLOR DISPLAY PANEL-MIPI

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1 Scope

This Specification defines AMOLED manufactured by EverDisplay Optronics(Shanghai) Limited, from here on refer as EDO. In the case of any unspecified item, it may require both EDO and the party designs this module into its product to work out a solution.

2 Features

2.1 Product Applications

Smart Watch

2.2 Product Features

- 1) Display color: 16.7M (RGB x 8bits)
- 2) Display format: 1.4"(400RGBx400)
- 3) Pixel arrangement: Real RGB arrangement
- 4) Interface: MIPI 1-lane
- 5) Driver IC: RM67162

3 Mechanical Specifications

Item	Specification	unit
Dimension outline	38.6 x 40.5 x 0.69	mm
LTPS Glass outline	38.6 x 40.5	mm
Encapsulation Glass outline	φ38.6	mm
Number of dots	400(W) x RGB x 400(H)	dots
Active area	φ35.4	mm
Diagonal size	1.39	inch
Pixel pitch	29.5 x 88.5	μm
Glass thickness (LTPS/encapsulation glass)	0.2 / 0.2	mm
Weight	2	g

Note: Refer to 9 Outline Dimension Drawing

4 Maximum Rating

Parameter	Symbol	Spec			Unit	Note
		Min.	Typ.	Max.		
Analog/boost power voltage	VCI	-0.3	-	5.5	V	-
I/O voltage	VDDIO	-0.3	-	5.5	V	-
Operating temperature	Top	-20	-	60	℃	-
Storage temperature	Tstg	-30	-	70	℃	-

5 Electrical Specifications

5.1 Electrical Characteristics

5.1.1 Power Characteristic:

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
AMOLED Power positive	ELVDD	4.55	4.6	4.65	V	-
AMOLED power Negative	ELVSS	-2.45	-2.4	-2.35	V	Ref
Digital Power supply	VDDIO	1.65	1.8	1.95	V	Ref
Analog Power supply	VCI	2.7	2.8	2.9	V	Ref

1) Normal Mode

Power Supply: IOVCC=1.8V VCI=2.8V

Frame Frequency: F_{frame}=60HZ @ 25degC, Brightness 350 nits, Command Mode

Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
100% Pixel On 350nits	IELVDD /ELVSS	-	20	24	mA	Ref
	IVCI	-	7	8.4	mA	Ref
	IVDDIO	-	2.0	2.4	mA	Ref

2) Idle Mode

Power Supply: IOVCC=1.8V VCI=2.8V

Frame Frequency: F_{frame}=15HZ @ 25degC, Brightness 30 nits,OPR:10%,Command Mode

Entry instruction: ->Reg:0xFE Para:0x00 ->Reg:0x39 Para:0x00

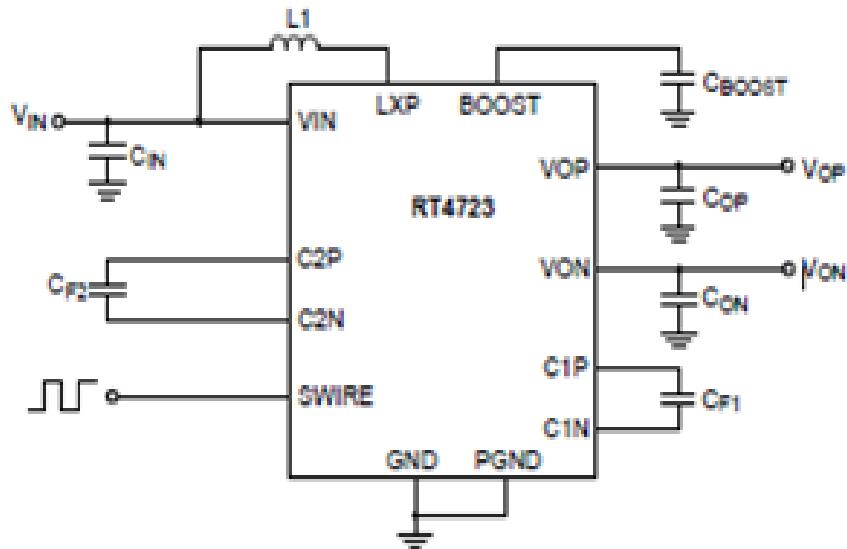
Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
10% Pixel On 30 nits	IELVDD /ELVSS	-	-	-	mA	Supplied by Driver IC
	IVCI	-	3	3.6	mA	Ref
	IVDDIO	-	1	1.2	mA	Ref

3) Deep Standby Mode

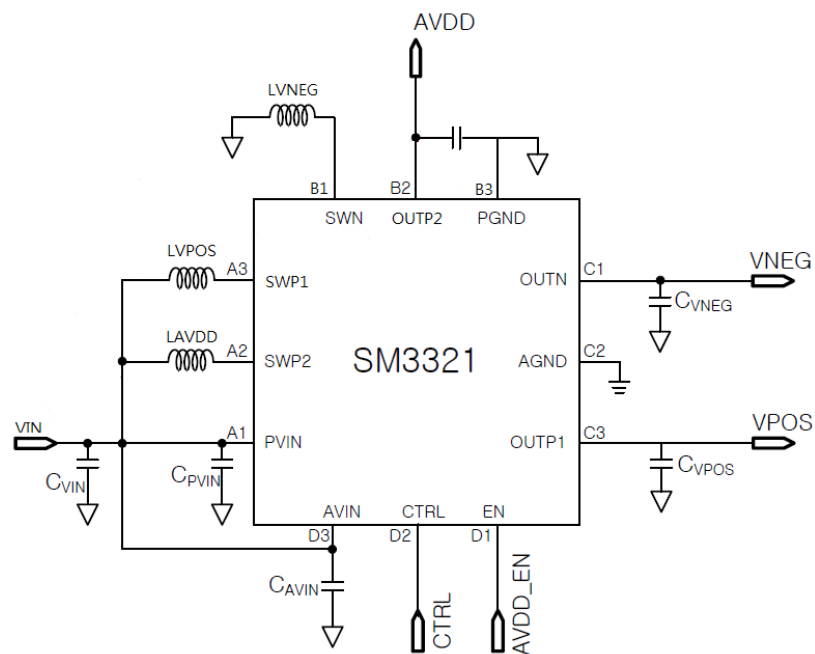
Display Condition	Symbol	Min.	Typ.	Max.	Unit	Remark
Deep Standby	IVCI	-	-	3	uA	-
	IVDDIO	-	-	1	uA	-

5.1.2 Power supply circuit application (This is for reference only):

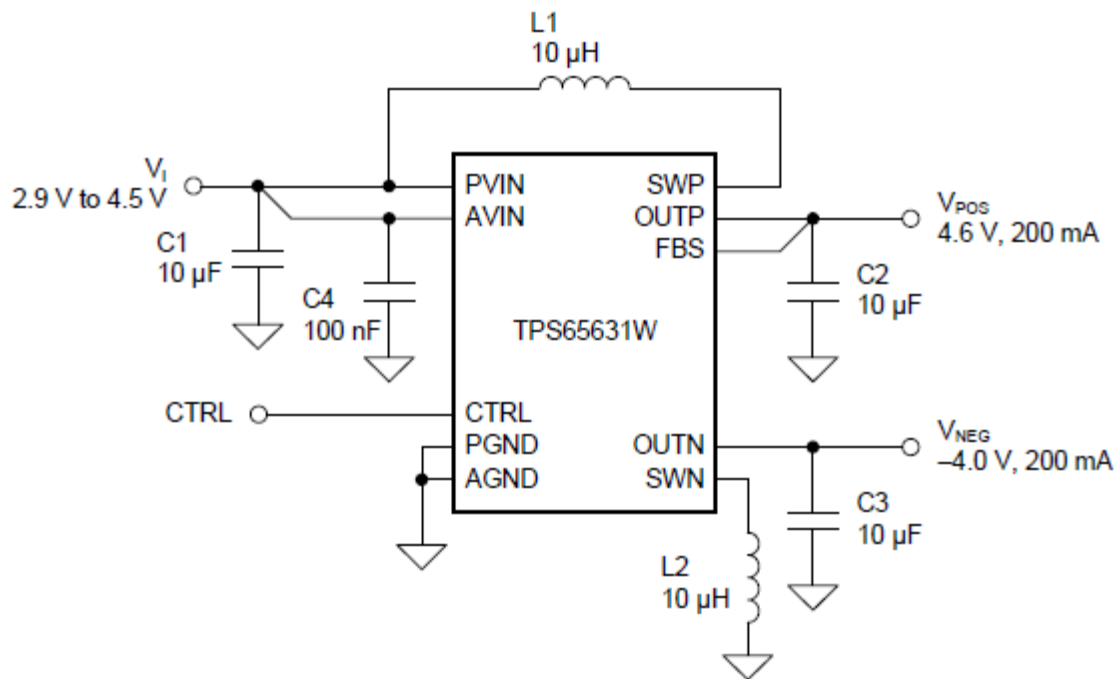
1) RT4723



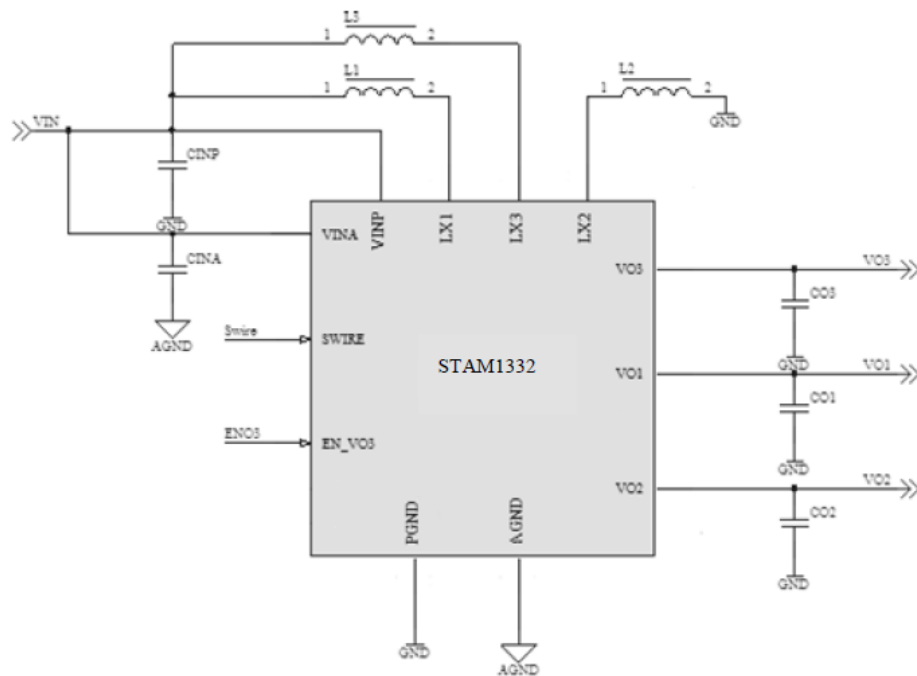
2) SM3321



3) TPS65631W



4) STAM1332



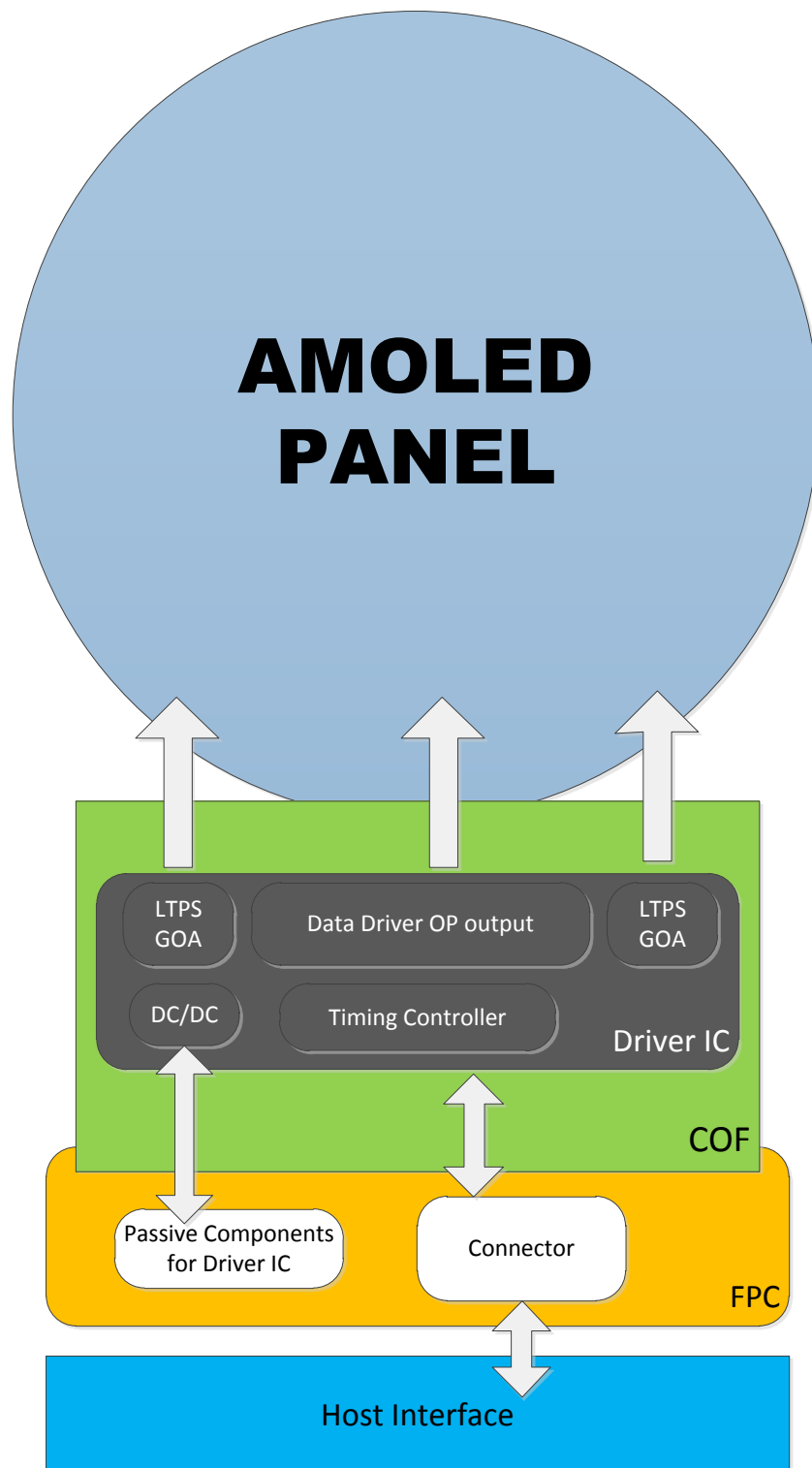
5.2 I/O Connection and Block Diagrams

5.2.1 I/O Connection

#	Pin_name	I/O	Description
1	GND	Power	The power ground
2	XRES	I	Device reset signal (0:enable;1:disable)
3	DSI_D0N	I/O	MIPI DSI data0-
4	SWIRE	O	Swire protocol setting pin of Power IC
5	DSI_D0P	I/O	MIPI DSI data0+
6	NC(MTP)	-	Floating
7	GND	Power	The power ground
8	TE	O	Tear effect output
9	DSI_CLKN	I/O	MIPI DSI clock-
10	GND	Power	The power ground
11	DSI_CLKP	I/O	MIPI DSI clock+
12	GND	Power	The power ground
13	GND	Power	The power ground
14	GND	Power	The power ground
15	VDDIO	Power	Driver IC digital I/O supply
16	VCI	Power	Driver IC analog supply
17	GND	Power	The power ground
18	GND	Power	The power ground
19	ELVSS	Power	AMOLED power Negative
20	ELVDD	Power	AMOLED power Positive
21	ELVSS	Power	AMOLED power Negative
22	ELVDD	Power	AMOLED power Positive
23	ELVSS	Power	AMOLED power Negative
24	ELVDD	Power	AMOLED power Positive

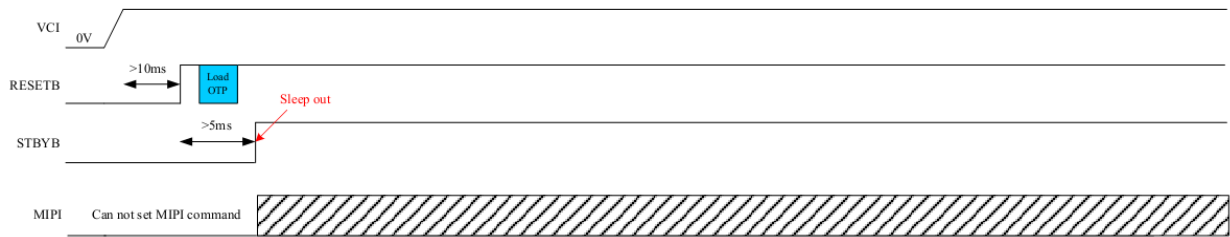
5.3 Initial CODE

TBD

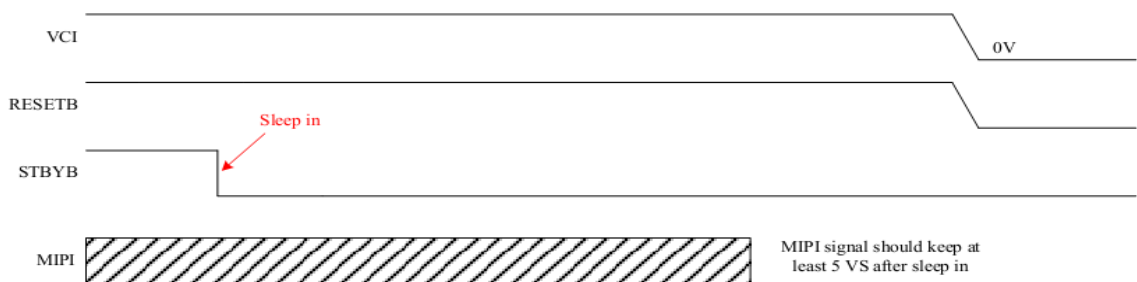


5.4 Recommended Operating Sequence

5.4.1 Power on sequence

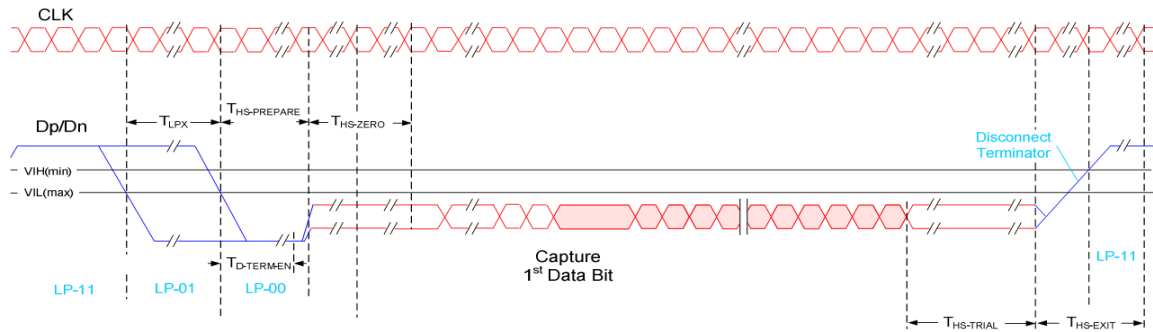


5.4.2 Power off sequence

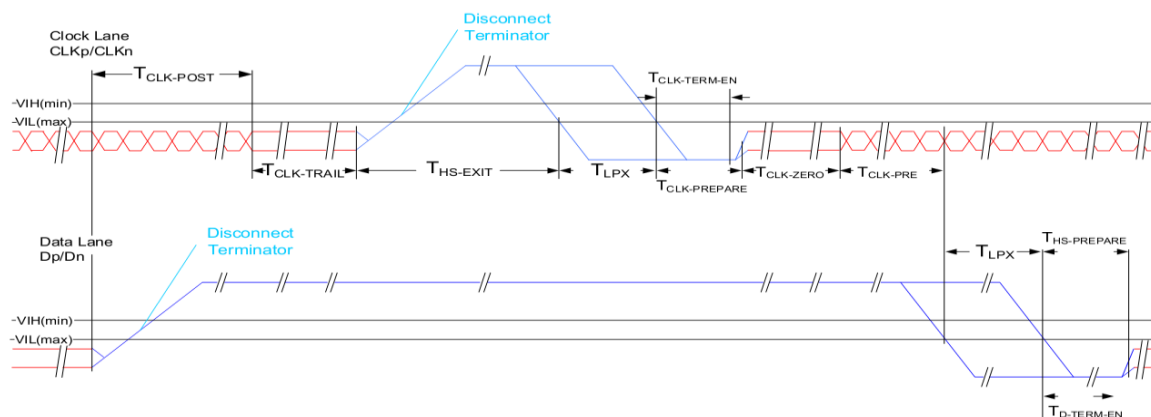


5.5 AC Characteristics (MIPI)

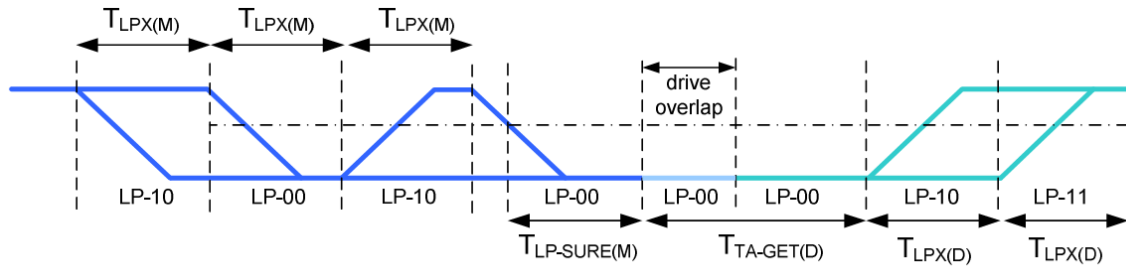
5.5.1 HS Data Transmission Burst



5.5.2 HS Clock Transmission



5.5.3 Turnaround Procedure



5.5.4 Timing Parameters

Symbol	Description	Min	Typ	Max	Unit
TREOT	30%-85% rise time and fall time	-	-	35	ns
TCLK-MISS	Timeout for receiver to detect absence of Clock transitions and disable the Clock Lane HS-RX.	-	-	60	ns
TCLK-POST*1	Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of THS-TRAIL to the beginning of TCLK-TRAIL.	60ns + 52*UI (For DCS)	-	-	ns
TCLK-PRE	Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.	8	-	-	ns
TCLK-SETTLE	Time interval during which the HS receiver shall ignore any Clock Lane HS transitions, starting from the beginning of TCLK-PRE.	95	-	300	ns
TCLK-TERM-EN	Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses VIL,MAX.	Time for Dn to reach VTERM-EN	-	38	ns
THS-SETTLE	Time interval during which the HS receiver shall ignore any Data Lane HS transitions, starting from	85 ns + 6*UI	-	145 ns + 10*UI	ns

	the beginning of THSPREPARE.				
TEOT	Time from start of THS-TRAIL or TCLK-TRAIL period to start of LP-11 state	-	-	105ns+48*UI	ns
THS-EXIT(1)	time to drive LP-11 after HS burst	100	-	-	ns
THS-PREPARE	Time to drive LP-00 to prepare for HS transmission	40ns + 4*UI	-	85ns+6*UI	ns
THS-PREPARE + THS-ZERO	THS-PREPARE + Time to drive HS-0 before the Sync sequence	145ns + 10*UI	-	-	ns
THS-SKIP	Time-out at RX to ignore transition period of EoT	40	-	55ns+4*UI	ns
THS-TRAIL	Time to drive flipped differential state after last payload data bit of a HS transmission burst	60 + 4*UI	-	-	ns
TLPX	Length of any Low-Power state period	50	-	-	ns
Ratio TLPX	Ratio of TLPX(MASTER)/TLPS(SLAVE) between Master and Slave side	2/3	-	3/2	ns
TTA-GET	Time to drive LP-00 by new TX	5*TLPX	5*TLPX	5*TLPX	ns
TTA-GO	Time to drive LP-00 after Turnaround Request	4*TLPX	4*TLPX	4*TLPX	ns
TTA-SURE	Time-out before new TX side starts driving	TLPX	-	2*TLPX	ns

5.5.5 Timing requirements for RESETB

When RESETB of the reset pin equals to Low, it will be in the condition of reset.

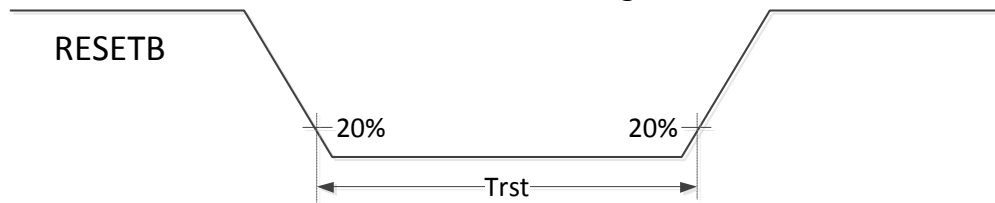
When it is in the condition of reset, it will make the device recover the initial set.

However, in order to avoid the reset noise cause reset, there is a mechanism to judge about whether the reset is needed or not.

The closed interval of Low can be shown as the following.

(Test condition: VDDIO=1.65V~3.6V, VSS=0V, TA=-20°C~+70°C)

Parameter	Symbol	Conditions	Spec			Unit
			Min.	Typ.	Max.	
Reset low pulse width	Trst	-	20	-	-	μs

Table: Reset timing

Figure: Reset timing

6 Electro-Optical Specification

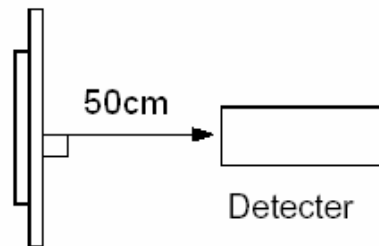
Test condition: IOVCC=1.8V , VCI=2.8V ,Ta=25℃

Item		Symbol	Condition	Value			Unit	Note
				Min	Typ	Max		
Luminance			$\theta=0^{\circ}$ $\Phi=0^{\circ}$	315	350	385	cd/m2	Note 1
Uniformity				85		-	%	Note 2
Viewing Angle	Left	θ_L	$Cr\geq 200$	80	85	-	Deg.	Note 3
	Right	θ_R		80	85	-		
	Top	ψ_T		80	85	-		
	Bottom	ψ_B		80	85	-		
Contrast Ratio		CR	$\theta=0^{\circ}$ $\Phi=0^{\circ}$	5000	10000	-	-	Note 4
Response Time		Tr+Tf		-	-	2	ms	Note 5
Color Coordinate of CIE1931	Red	X	$\theta=0^{\circ}$ $\Phi=0^{\circ}$	0.64	0.67	0.70	-	-
		Y		0.30	0.33	0.36		
	Green	X		0.16	0.21	0.26		
		Y		0.68	0.73	0.78		
	Blue	X		0.09	0.13	0.17		
		Y		0.02	0.06	0.10		
	White	X		0.28	0.30	0.32		
		Y		0.29	0.31	0.33		
NTSC Ratio		NTSC	CIE1931	85	100	-	%	-
Flicker		-	-	-	-	-30	dB	-
Gamma		-	-	1.9	2.2	2.5		Note 6
Crosstalk		ΔCT	-	-	-	1.1		Note 7

Note 1: Luminance measurement

The test condition is measured on the surface of AMOLED module at 25℃.

- Measurement equipment CS2000 or similar equipment (Field of view:1deg,Distance:50cm)
- Measuring surroundings: Dark room.
- Measuring temperature: Ta=25℃.
- Adjust operating voltage to get optimum contrast at the center of the display.



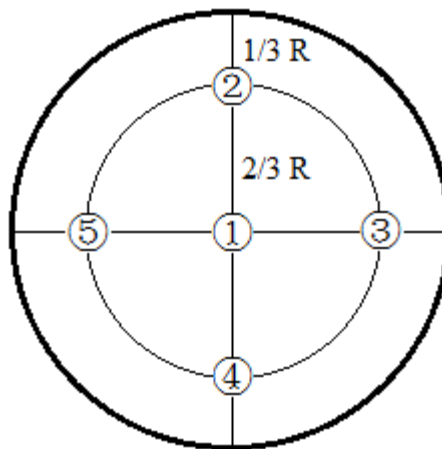
Note 2: Uniformity

The luminance uniformity is calculated by using following formula:

$$\Delta Bp = Bp (\text{Min.}) / Bp (\text{Max.}) \times 100 (\%)$$

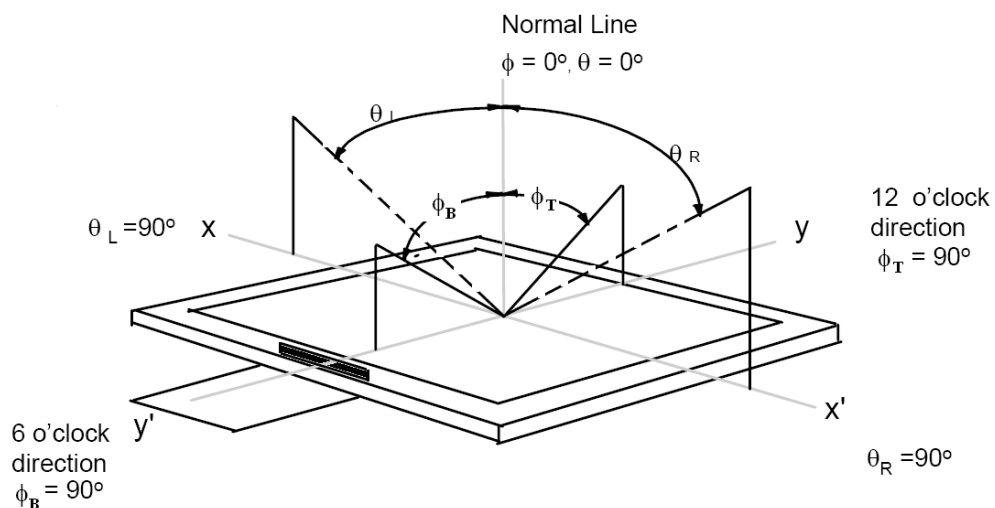
Bp (Max.) = Maximum brightness in 5 measured spots

Bp (Min.) = Minimum brightness in 5 measured spots.



Note 3: The definition of Viewing Angle

Refer to the graph below marked by ϑ and ϕ

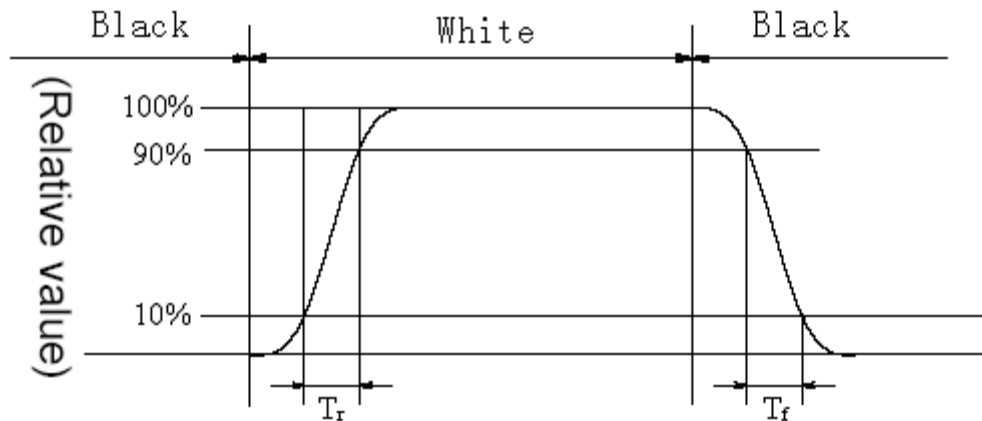


Note 4: The definition of Contrast Ratio:

$$\text{Contrast Ratio (CR)} = \frac{\text{Luminance When AMOLED is at "White" state}}{\text{Luminance When AMOLED is at "Black" state}}$$

Note 5: Definition of Response time.

The output signals of photo detector are measured when the input signals are changed from “black” to “white” (Voltage falling time) and from “white” to “black” (Voltage rising time), respectively. The response time is defined as the time interval between the 10% and 90% of amplitudes. Refer to figure as below.


Note 6: Gamma curve

The whole curve's tolerance must control within +/-0.3, test the gray scale below:

8, 16, 25, 33, 41, 49, 58, 66, 74, 82, 90, 99, 107, 115, 123, 132, 140, 148, 156, 165, 173, 181, 189, 197, 206, 214, 222, 230, 239, 255

Note 7: Crosstalk

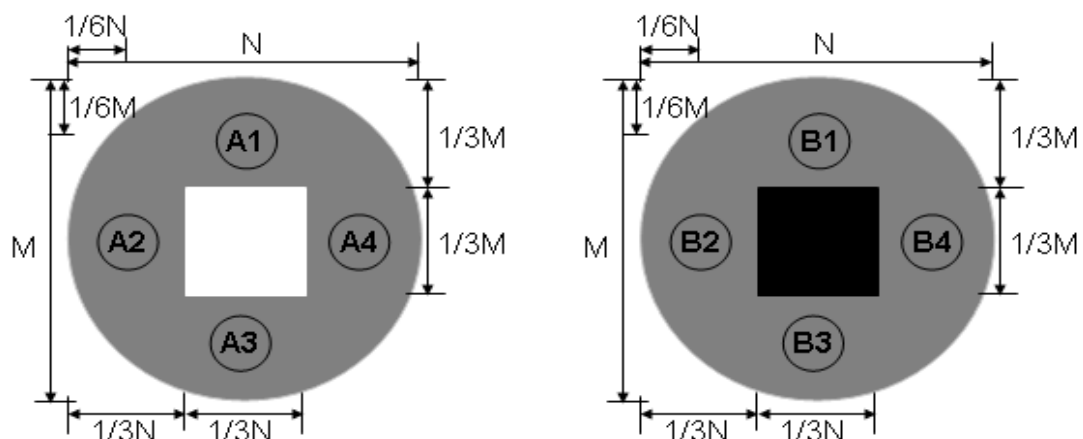
There should be no visible cross-talk in normal direction of the display when the two “Cross-talk Test Patterns” below are loaded.

ΔBp (Max.) = Maximum value in $\Delta Bp1 \sim \Delta Bp4$.

ΔBp (Min.) = Minimum value in $\Delta Bp1 \sim \Delta Bp4$.

$\Delta CT = \Delta Bp$ (Max.) / ΔBp (Min.).

ΔCT must be less than 1.10



Cross-talk Test Pattern

7 Reliability

7.1 Environmental Test

Item	Main spec	No. of failures / No. of examinations	Note
High Temperature Operation	70℃/ 128hours	0/5	
Low Temperature Operation	-20℃/128hours	0/5	
High Temperature Storage	80℃/128hours	0/5	
Low Temperature Storage	-40℃/ 128hours	0/5	
High Temperature Humidity Operation	60℃/90%RH 128hours	0/5	
Thermal Shock	-40℃~80℃ 0.5hr, 100 cycles	0/5	

7.2 Electrical Test

Item	Main spec	Note
Air Discharge	±4kV , 150pF/330Ω (B class) ±6kV , 150pF/330Ω (C class @machine test)	5Points, Each 2times. No degradation of OLED performance after this test.
Contact Discharge	±4kV, 150pF/330Ω (Module level)	B class: Sample在实施静电放电时，画面有横线，闪烁，画异,无显等异常，但在不实施静电放电时恢复正常显示. C class: Sample在实施静电放电时，画面有横线，闪烁，画异,无显等异常，且在不实施静电放电时不能恢复，但信号Reset后能恢复正常显示.

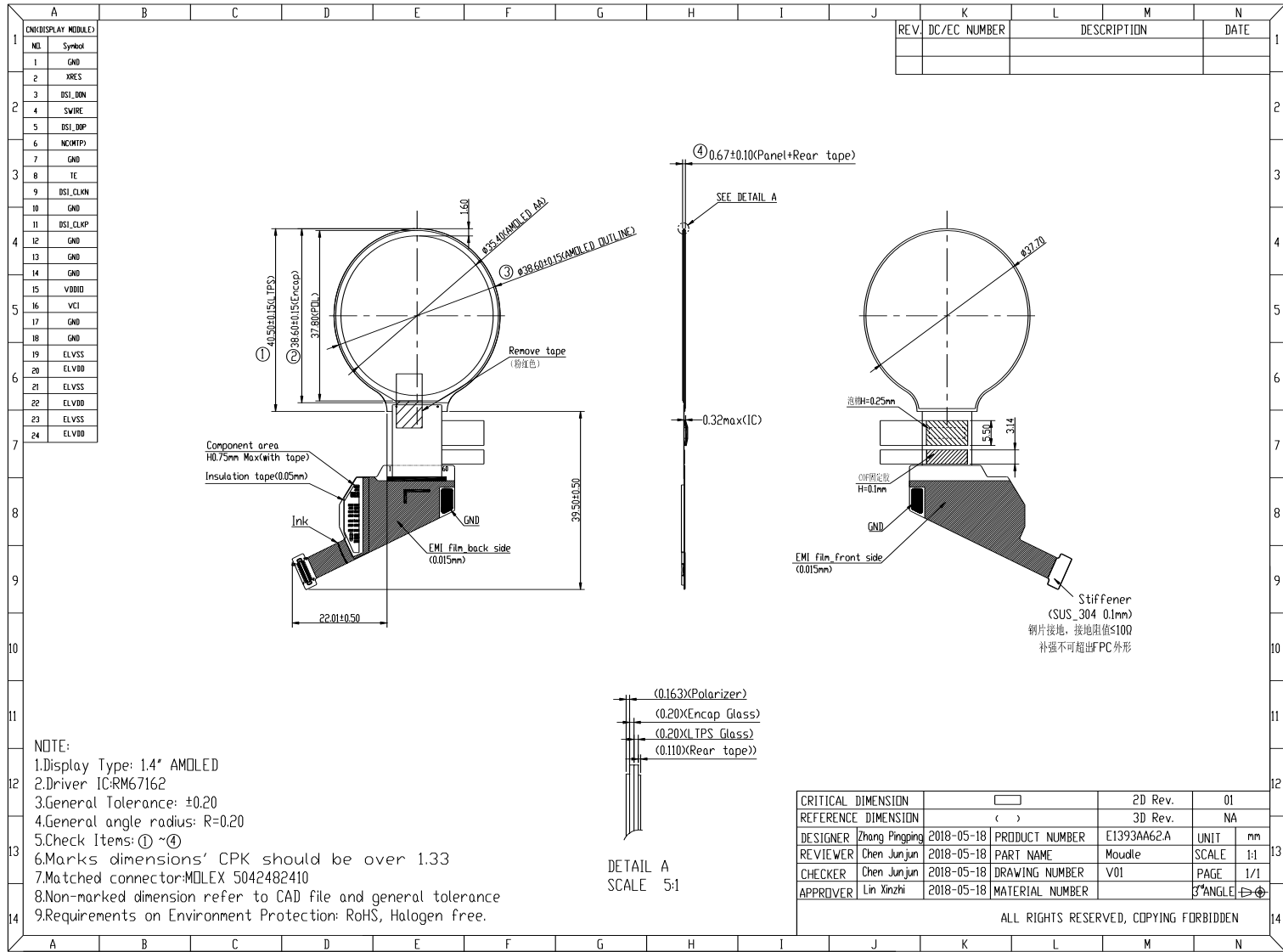
7.3 Mechanical Test

Item	Main spec	Note
Drop Test	Drop the packing from 1.0m height, 3 times for 6-faces, 3-edges and 1-corner	Package
Vibration-proof test	2g, f=10->55->10Hz apply in each of X, Y, and Z direction for 30 min	Package
Glass Strength Test -4PB	B10≥20N	Panel 80 pcs
FPC peeling	Horizontal: 10N/cm Vertical: 5N/cm	Panel 10pcs

8 Handling Precautions

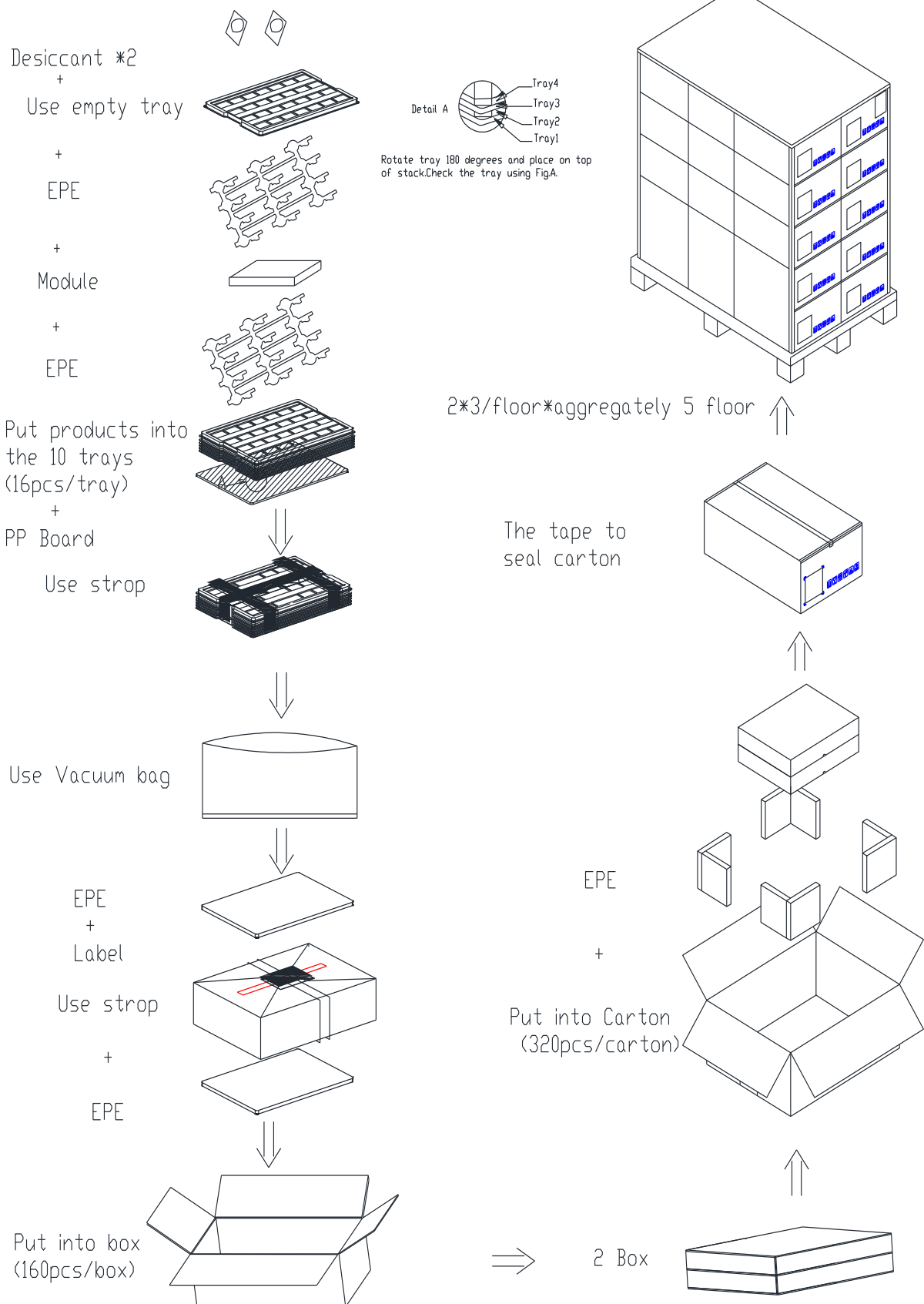
- 8.1 When cleaning ITO pad, avoid using hard and abrasive material or corrosive solution
- 8.2 Keep module away from direct sunlight or fluorescent light, and keep it at room temperature and humidity
- 8.3 Strong impact & pressure on module and packing is prohibited
- 8.4 Following normal power on/off sequence is necessary for preventing abnormal display or permanent damage to display
- 8.5 Optimal contrast ratio under ideal voltage is AMOLED module's characteristic, hence it is recommended a voltage control function available
- 8.6 Image sticking may occur if an image displays for an extended period of time
- 8.7 When interfered by system's overall mechanical design, an abnormal display may occur
- 8.8 After considering emitting energy, you should plan your design to satisfy EMI standards.
- 8.9 Host side should place a surge-prevent circuit at power trace (ie: VCI, Vddi) to protect AMOLED module.

9 Outline Dimension Drawing



10 Packing Specification

(1)



(2) Inner packing (material/identifying/package quantity)

NO	Item	Material	Size	Quantity	Module Quantity	Note
1	Box	paper	516*338*248	1	320	
2	Tray	PET	455*290*14	22	16	
3	Vacuum bag	Al	440*660*0.28	2	/	
4	EPE	EPE	408.96*244*1	40	/	
5	EPE2	EPE2	457*292*10	4	/	
6	Label of Inner Box	PET	52*100*0.075	2	/	
7	PP Board	PET	457*292*5	2	/	

(3) Outer packing (material/identifying/package quantity)

NO	Item	Material	Size	Quantity	Module Quantity	Note
1	Carton	paper	459*294*115	2	160	
2	Label of outer Box	paster	52*100*0.1	1	/	
3	EPE bead	EPE	120*244*100	4	/	

11 The Control of Hazardous substances

The Control of Hazardous substances refer to EDO document 《有害物质管控标准书》
(Standard document for the Control of Hazardous substances) EDO-IS-110, Version05